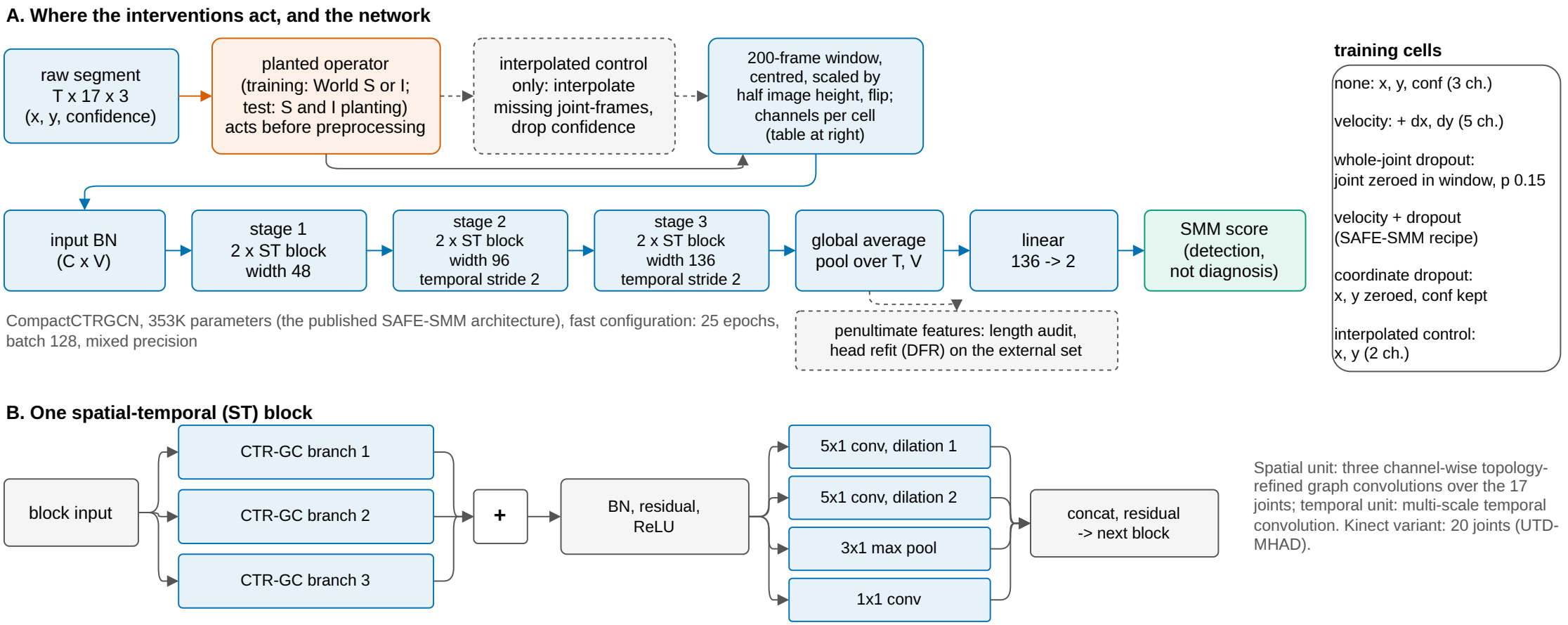
A. Where the interventions act, and the network
raw segment
T x 17 x 3
(x, y, confidence)
planted operator
(training: World S or I;
test: S and I planting)
acts before preprocessing
interpolated control
only: interpolate
missing joint-frames,
drop confidence
200-frame window,
centred, scaled by
half image height, flip;
channels per cell
(table at right)
input BN
(C x V)
stage 1
2 x ST block
width 48
stage 2
2 x ST block
width 96
temporal stride 2
stage 3
2 x ST block
width 136
temporal stride 2
global average
pool over T, V
linear
136 -> 2
SMM score
(detection,
not diagnosis)
CompactCTRGCN, 353K parameters (the published SAFE-SMM architecture), fast configuration: 25 epochs, batch 128, mixed precision
penultimate features: length audit,
head refit (DFR) on the external set
training cells
none: x, y, conf (3 ch.)
velocity: + dx, dy (5 ch.)
whole-joint dropout:
joint zeroed in window, p 0.15
velocity + dropout
(SAFE-SMM recipe)
coordinate dropout:
x, y zeroed, conf kept
interpolated control:
x, y (2 ch.)
B. One spatial-temporal (ST) block
block input
CTR-GC branch 1
CTR-GC branch 2
CTR-GC branch 3
+
BN, residual,
ReLU
5x1 conv, dilation 1
5x1 conv, dilation 2
3x1 max pool
1x1 conv
concat, residual
-> next block
Spatial unit: three channel-wise topology-refined graph convolutions over the 17 joints; temporal unit: multi-scale temporal convolution. Kinect variant: 20 joints (UTD-MHAD).